Subject-Excluded Prototype Contrastive Learning for Cross-Subject SEED-IV EEG Emotion Recognition

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Abstract

EEG-based emotion recognition is a promising direction for affective brain–computer interfaces, but cross-subject generalization remains difficult because EEG recordings vary substantially across individuals. We study subject-aware representation learning for four-class SEED-IV emotion recognition using pre-extracted differential entropy features and a compact EEGNet-style spatial–spectral backbone. We propose Subject-Excluded Prototype Contrastive learning (SEPC), a prototype-based contrastive objective that constructs class prototypes for each anchor using only subjects other than the anchor's subject. Under leave-one-subject-out evaluation on all 15 SEED-IV subjects, SEPC outperforms the global prototype baseline on 10 of 15 held-out subjects, improving mean accuracy from 42.6% to 43.2%. This confirms that subject exclusion provides a consistent inductive bias for prototype-based cross-subject learning. However, both prototype methods trail cross-entropy (44.7%) and supervised contrastive (44.6%) baselines under the tested configuration, indicating that the prototype framework requires stronger regularization or larger backbones to realize its full potential.

1 Introduction

Electroencephalography (EEG) is widely used in brain–computer interface research because it is non-invasive, relatively inexpensive, and has high temporal resolution. These properties make EEG attractive for affective computing, where the goal is to infer a user's emotional state from neural recordings (2; 3; 4). Reliable EEG emotion recognition could support affect-aware interfaces, adaptive learning systems, and human–computer interaction systems that respond to users' cognitive or emotional state.

A central challenge is cross-subject generalization. EEG signals vary across individuals due to differences in physiology, cortical structure, electrode placement, baseline activity, and recording noise. Models trained and tested on data from the same subject can exploit subject-specific regularities that may not transfer to unseen users. This limits practical deployment: an affective BCI system should ideally work for a new user without requiring extensive subject-specific retraining.

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Contrastive learning offers a natural way to address this problem because it shapes the representation space directly. Supervised contrastive learning pulls together embeddings with the same label and separates embeddings with different labels (9). Prior work has explored contrastive learning for subject-robust EEG emotion recognition (10; 11). However, standard supervised contrastive learning and ordinary prototype contrast do not explicitly distinguish between positive signals coming from the current subject and positive signals coming from other subjects.

This distinction matters in cross-subject evaluation. If a class prototype is computed using all source subjects, then for a training example from subject s_i, the positive prototype partially contains information from s_i itself. This creates a weaker cross-subject training signal: the anchor can be pulled toward a target partly supported by its own subject-specific structure. A stricter cross-subject target would define the positive class prototype using only other subjects.

We propose Subject-Excluded Prototype Contrastive learning (SEPC), a simple loss-level modification for cross-subject EEG emotion recognition. For an example from subject s_i and emotion class y_i, the positive prototype for class y_i is computed using only subjects $s \neq s_i$. Negative prototypes are constructed in the same leave-subject-out manner. We use a compact EEGNet-style backbone to isolate the effect of the training objective (1).

Contributions.

1. We formulate Subject-Excluded Prototype Contrastive learning for four-class cross-subject SEED-IV EEG emotion recognition.
2. We adapt an EEGNet-style model to pre-extracted SEED-IV DE/LDS features, treating each input as a spatial–spectral matrix of shape 62×5.
3. We evaluate four training objectives—cross-entropy, supervised contrastive, global prototype contrast, and SEPC—under full 15-fold leave-one-subject-out evaluation.
4. We show that subject exclusion consistently improves prototype-based learning (10/15 subjects), isolating the value of this inductive bias.

2 Related Work

EEG emotion recognition and SEED-IV. SEED-IV is a four-emotion EEG dataset with labels neutral, sad, fear, and happy (4). The dataset includes three sessions per subject, each with 24 trials and a session-specific label sequence. Because data are collected from multiple subjects, SEED-IV is suitable for studying cross-subject emotion recognition. Evaluation protocol is critical: random sample-level splits overestimate performance by allowing subject-specific structure to leak across splits.

Compact EEG models. EEGNet is a compact convolutional architecture designed for EEG-based BCI tasks (1). It captures channel and feature interactions with relatively few parameters. Since our inputs are pre-extracted differential entropy (DE) features rather than raw EEG time series, we describe our model as an EEGNet-style spatial–spectral network.

Cross-subject generalization. Cross-subject EEG emotion recognition has motivated domain adaptation (5), adversarial alignment (6), graph-based modeling (7), and plug-and-play adaptation (8). SEPC is simpler than these frameworks: it does not introduce an explicit adversary or adaptation module, instead treating each subject as a source domain when constructing contrastive prototypes.

Contrastive and prototype learning. Supervised contrastive learning encourages class-structured representation spaces (9). Contrastive methods have been applied to EEG emotion recognition for subject-robust representations (10; 11). Prototype-based objectives summarize each class by representative embeddings. SEPC differs from ordinary global prototype contrast in that the target prototype for each example excludes the example's own subject.

3 Method

3.1 Problem Setup

Let $x_i \in \mathbb{R}^{62\times 5}$ denote a pre-extracted SEED-IV DE/LDS feature window, $y_i \in \{0,1,2,3\}$ its emotion label, and $s_i \in \{1,\ldots,S\}$ its subject identity. The encoder maps a feature window to a hidden representation $h_i = f_\theta(x_i)$, and a classifier head predicts the emotion label via $q_\psi(y_i \mid h_i)$. The standard cross-entropy objective is $\mathcal{L}_{\mathrm{CE}} = -\frac{1}{N}\sum_{i=1}^{N} \log q_\psi(y_i \mid h_i)$.

3.2 Spatial–Spectral Backbone

Each SEED-IV trial feature array has shape $(62, T, 5)$, where 62 is the number of EEG channels, T is the number of 4-second windows, and 5 is the number of frequency bands (delta, theta, alpha, beta, gamma). Each window yields one sample $x_i = \mathrm{arr}[:, t, :] \in \mathbb{R}^{62\times 5}$, which becomes $\mathbb{R}^{1\times 62\times 5}$ after adding the convolutional channel dimension. The first convolution operates along the 5-band spectral axis; a depthwise spatial convolution then mixes electrodes for each learned spectral filter. A projection head maps the encoder output h_i to a 64-dimensional normalized embedding $z_i = u_i/\|u_i\|_2$ for contrastive losses, while a separate classification head maps h_i to emotion logits.

3.3 Subject-Class Prototypes

Both the global prototype baseline and SEPC maintain a prototype $P_{s,c} \in \mathbb{R}^d$ for each subject s and class c, updated by exponential moving average:

$$P_{s,c} \leftarrow \alpha P_{s,c} + (1-\alpha)\,\mathrm{mean}\{z_i : s_i = s,\ y_i = c\}.$$

The prototype is normalized after each update. Let $m_{s,c} = 1$ indicate that $P_{s,c}$ has been initialized. The two methods differ only in how per-subject-class prototypes are aggregated into class-level targets.

3.4 Subject-Excluded Prototype Contrastive Learning

For sample i from subject s_i, SEPC constructs the class-c prototype by averaging over subjects other than s_i:

$$p_c^{(-s_i)} = \mathrm{normalize}\left(\frac{\sum_{s\neq s_i} m_{s,c}\,P_{s,c}}{\sum_{s\neq s_i} m_{s,c}}\right).$$

The SEPC loss for example i is

$$\mathcal{L}_{\mathrm{SEPC}}(i) = -\log \frac{\exp(z_i^\top p_{y_i}^{(-s_i)}/\tau)}{\sum_{c=0}^{3}\exp(z_i^\top p_c^{(-s_i)}/\tau)},$$

where $\tau > 0$ is a temperature parameter. The full objective is $\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \lambda \cdot \frac{1}{|\mathcal{I}|}\sum_{i\in\mathcal{I}} \mathcal{L}_{\mathrm{SEPC}}(i)$, where $\mathcal{I}$ is the set of samples with valid subject-excluded prototypes.

3.5 Motivation: Why Subject-Excluded Prototypes?

A global class prototype averages over all source subjects: $p_c^{\mathrm{glob}} = \frac{1}{|\mathcal{S}_c|}\sum_{s\in\mathcal{S}_c} P_{s,c}$. For an anchor from subject s_i, this prototype includes $P_{s_i,c}$ with weight $1/|\mathcal{S}_c|$. Under a stylized additive decomposition $z_i \approx \mu_{y_i} + b_{s_i} + \epsilon_i$ (where μ_{y_i} is emotion structure and b_{s_i} is subject-specific bias), the global prototype contains the current subject's bias at weight $1/S$, while the SEPC prototype assigns it weight zero.

Observation 1. *Under the additive model, a balanced global class prototype includes the current subject's component with weight $1/S$; the SEPC prototype assigns it weight zero.*

This is not a proof of subject invariance—real EEG embeddings need not decompose additively. Rather, it identifies the inductive bias: SEPC ensures that the contrastive target for each example is supported entirely by other subjects.

3.6 Comparison Objectives

We compare SEPC against three baselines: (1) **CE**: cross-entropy only; (2) **SupCon**: cross-entropy plus supervised contrastive loss (9); (3) **Prototype**: global prototype contrast using the same EMA infrastructure as SEPC but including the anchor subject in prototype aggregation. The Prototype baseline is the most important comparison because it isolates the effect of subject exclusion.

4 Experiments

4.1 Dataset and Evaluation Protocol

We use SEED-IV (4), a four-class EEG emotion dataset with 15 subjects, three sessions per subject, and 24 trials per session. We use pre-extracted DE/LDS features from the `eeg_feature_smooth` files, where each trial has shape $(62, T, 5)$. Each subject contributes 2,505 windows, yielding 37,575 total. We perform full leave-one-subject-out (LOO) evaluation: in each of 15 folds, one subject is held out (2,505 windows) and the model is trained on the remaining 14 subjects (35,070 windows). The held-out subject is never used for training, prototype construction, or normalization fitting.

4.2 Implementation Details

All four methods use the same backbone with input shape $(B, 1, 62, 5)$: 8 spectral filters, depth multiplier 2, producing 16 depthwise spatial filters. The projection head outputs 64-dimensional embeddings. Training uses Adam (lr $= 10^{-3}$, weight decay $= 10^{-4}$), balanced batches of 32 (8 per class), and z-score normalization per electrode and frequency band (fit on training data only). Augmentation applies Gaussian noise ($\sigma = 0.05$) and random frequency-band dropout ($p = 0.2$). All runs train for 50 epochs with gradient clipping at norm 1.0 and fixed random seeds.

For contrastive and prototype methods, a 5-epoch warmup trains only the contrastive/prototype loss before adding classification loss. After warmup, the combined objective uses $\lambda = 0.5$, $\tau = 0.1$, and EMA momentum $\alpha = 0.9$. CE training uses no warmup. All hyperparameters are fixed across folds. Each LOO fold trains in approximately 4–6 minutes on a single NVIDIA RTX 2080 Ti.

4.3 Results

Aggregate performance. Table 1 summarizes the LOO results. CE achieves the highest mean accuracy at 44.7%, with SupCon essentially tied at 44.6%. Both prototype-based methods trail: SEPC reaches 43.2% and global Prototype reaches 42.6%. Figure 1 visualizes these results with standard deviation error bars. The four methods perform within a $\sim$2 percentage-point range, with all standard deviations near 9 pp, reflecting the large subject-to-subject variability inherent in cross-subject EEG evaluation.

Table 1: Cross-subject SEED-IV leave-one-subject-out results (15 folds). Best val accuracy is the peak validation accuracy across 50 epochs. Bold indicates the best mean.

Method	Mean (%)	Std (%)	Worst (%)	Best (%)
CE	**44.7**	9.3	31.0	63.9
SupCon	44.6	9.6	29.2	63.7
Prototype	42.6	8.5	28.6	56.7
SEPC	43.2	9.3	25.2	60.4

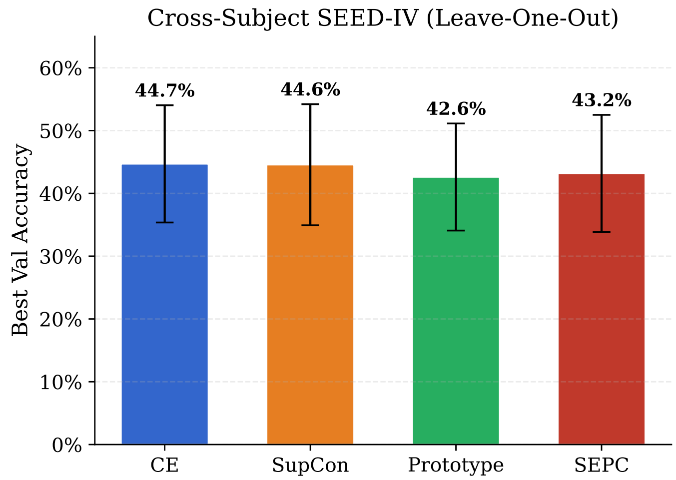

Figure 1: Mean best validation accuracy across 15 held-out subjects with standard deviation error bars. CE and SupCon are essentially tied; both prototype methods trail by 1–2 pp.

Subject-exclusion ablation. The most important comparison is SEPC versus global Prototype, since these share identical architecture, optimizer, hyperparameters, and EMA infrastructure and differ only in whether the anchor subject contributes to its own prototype. Figure 2 shows the per-subject accuracy difference. SEPC outperforms Prototype on 10 of 15 subjects, with a mean improvement of $+0.58$ pp. Gains appear on both easy subjects (subject 4: $+4.1$ pp; subject 6: $+6.9$ pp) and moderate subjects (subject 9: $+7.2$ pp; subject 11: $+2.4$ pp). The five subjects where Prototype wins include subject 8 (-9.7 pp), where exclusion may increase prototype estimation variance.

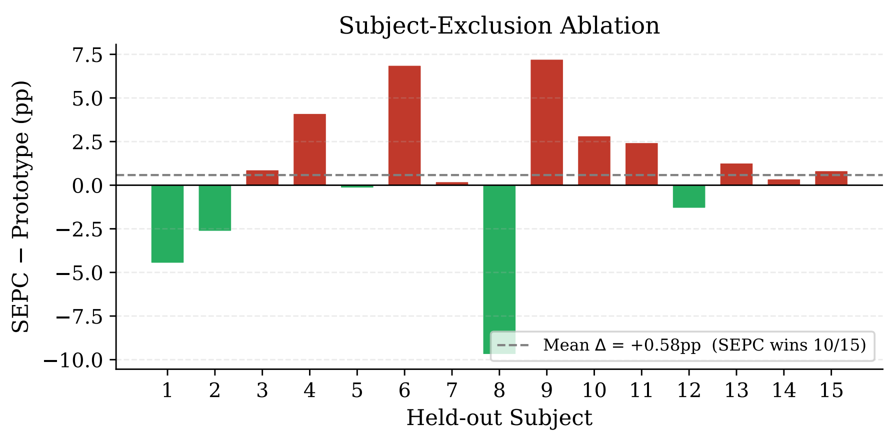

Figure 2: Per-subject accuracy difference (SEPC $-$ Prototype). Red bars: SEPC wins. Green bars: Prototype wins. SEPC improves on 10/15 subjects with a mean gain of $+0.58$ pp.

Per-subject breakdown. Table 2 and Figure 3 present the full breakdown. No single method dominates all subjects. Subject difficulty varies widely: subject 4 is easiest across all methods (56–64%) while subjects 8 and 11 are hardest (22–35%), a roughly twofold spread typical of cross-subject EEG evaluation.

Table 2: Per-subject best validation accuracy (%). Bold indicates the best method for each subject.

Method	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Mean
CE	40.6	39.0	**44.9**	**63.9**	**42.0**	**45.9**	61.7	34.6	36.9	44.1	**35.4**	44.1	31.0	50.5	**55.8**	44.7
SupCon	**41.4**	47.7	37.2	61.5	39.2	43.4	**63.7**	34.7	42.4	**47.5**	29.2	38.7	**35.4**	**56.8**	49.6	44.6
Prototype	41.2	**56.7**	36.4	56.2	40.9	38.4	46.9	**34.9**	36.9	40.6	28.6	**44.2**	31.5	53.7	51.9	42.6
SEPC	36.8	54.0	37.3	60.4	40.7	45.2	47.1	25.2	**44.1**	43.4	31.0	42.9	32.8	54.0	52.7	43.2

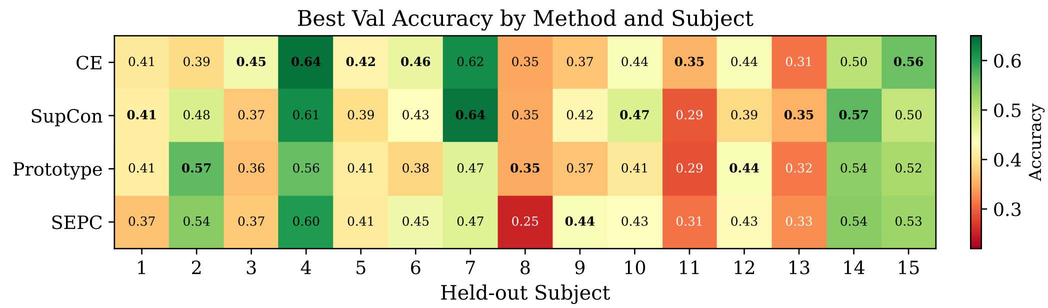

Figure 3: Best validation accuracy by method and held-out subject. Bold values indicate the per-subject best. Subject 4 is consistently easiest; subjects 8 and 11 are consistently hardest across all methods.

Overfitting and class collapse. All four methods exhibit substantial overfitting. Mean final training accuracy is approximately 61–63% while mean final validation accuracy is 30–35%, yielding a gap of roughly 28–30 pp. Best validation accuracy typically occurs within the first 10 epochs, after which validation performance degrades. A related pathology is class collapse: in many folds the model achieves 0.0 F1-score for one or more emotion classes, effectively predicting only a subset of labels. Fear is the most frequently collapsed class, followed by sad. This occurs across all four objectives and is therefore attributable to model capacity and data characteristics rather than loss function choice.

4.4 Reproducibility Details

All experiments are implemented in PyTorch and are reproducible through the submitted codebase. The repository contains separate source code, experiment scripts, configuration files, saved results, and a README with setup and run instructions. Random seeds are fixed for Python, NumPy, and PyTorch. The main scripts save training histories, model checkpoints, confusion matrices, and classification reports for each run.

5 Analysis

The results support two conclusions—one positive and one negative—which together characterize the value and limitations of SEPC.

Subject exclusion helps within the prototype framework. The primary ablation—SEPC versus global Prototype—is positive. SEPC wins on 10/15 subjects with a mean improvement of $+0.58$ pp. Because these methods share everything except the exclusion mechanism, this improvement is directly attributable to the subject-exclusion inductive bias. The consistency (10/15 rather than a few large wins) suggests that excluding the anchor subject from prototype construction produces a systematically better cross-subject training signal, as predicted by the motivation in Section 3.5. Among the five subjects where Prototype wins, subject 8 shows the largest margin (-9.7 pp). Subject 8 is one of the hardest across all methods (best accuracy 25–35%), suggesting that prototype estimation variance dominates the exclusion benefit when the base signal is weak.

Prototype methods do not outperform simpler baselines here. Both SEPC (43.2%) and Prototype (42.6%) trail CE (44.7%) and SupCon (44.6%). Several factors likely contribute. First, the severe overfitting (28–30 pp train–val gap) suggests the compact backbone overfits before the prototype loss can effectively shape representations. Second, prototype quality depends on batch diversity:

with balanced batches of 32 from 14 training subjects, some (subject, class) pairs may receive too few samples per batch for stable EMA estimation. Third, class collapse may interact with prototype learning—if the model stops predicting a class, that class's prototype becomes stale and provides no useful gradient signal.

Subject variability dominates method differences. The standard deviation across subjects ($\sim$9 pp) is roughly 15 times larger than the spread between the best and worst method means ($\sim$2 pp). Subject identity is a far stronger predictor of difficulty than choice of training objective. Subjects 4 and 7 are consistently easy, while subjects 8, 11, and 13 are consistently hard. This pattern is consistent with the known challenge of cross-subject EEG generalization and motivates worst-subject metrics alongside mean accuracy.

6 Discussion

SEPC introduces a simple cross-subject inductive bias into prototype contrastive learning by changing the source of the contrastive target rather than the model architecture. Our results confirm that this exclusion produces a consistent benefit within the prototype framework: SEPC improves over global Prototype on 10/15 subjects. However, the finding that both prototype methods trail CE and SupCon requires careful interpretation.

This gap does not imply that prototype-based cross-subject learning is fundamentally limited. Rather, the current configuration—a compact backbone, fixed hyperparameters, no early stopping, and no per-subject tuning—may not provide the conditions under which prototype objectives outperform simpler baselines. Prototype losses introduce additional sources of variance (EMA estimation, batch composition, prototype staleness) that may require larger models, stronger regularization, or adaptive schedules to manage effectively. The observation that best validation accuracy typically occurs within the first 10 epochs suggests that early stopping could substantially change method rankings.

The class collapse phenomenon—where the model assigns zero probability to one or more classes—affects all methods and warrants separate investigation. It may be addressable through focal loss, class-conditional augmentation, or curriculum strategies.

Limitations. SEPC is a loss-level inductive bias, not a guarantee of domain invariance; the encoder may still encode subject identity. The additive motivation in Section 3.5 is a simplified model. All results use a single hyperparameter configuration; a sweep over τ, λ, warmup, and learning rate might reveal settings where prototype methods close the gap with CE. The compact backbone may lack the capacity for prototype-based shaping to exceed direct classification. We do not apply early stopping, which likely penalizes all methods but may differentially affect those with different convergence dynamics. Finally, this study uses only SEED-IV with DE/LDS features; generalization to other datasets and feature types remains untested.

Future directions. The subject-exclusion principle is architecture-independent and could be applied to larger models (e.g., transformers), combined with domain-adaptation methods, or extended with per-subject prototype banks that persist across training. A hyperparameter sweep targeting prototype methods—particularly temperature and EMA momentum—is a natural next step.

7 Conclusion

We formulated Subject-Excluded Prototype Contrastive learning for cross-subject SEED-IV EEG emotion recognition and evaluated it under full 15-fold leave-one-subject-out evaluation. SEPC improved over global prototype contrast on 10 of 15 held-out subjects ($+0.58$ pp mean accuracy), confirming that removing the anchor subject's contribution to the prototype target provides a useful inductive bias. Both prototype methods trailed cross-entropy and supervised contrastive baselines, suggesting that the prototype framework requires stronger regularization or larger model capacity to outperform simpler objectives in this setting. The subject-exclusion principle is architecture-independent and could be integrated into larger models or domain-adaptation pipelines in future work.

7.1 Moonshot

Our extended work included building a model similar to CV-EEGNet's based on the paper reviewed by our team. During our method development, we were especially inspired by the model architecture and loss functions based on subject exclusion metrics from CV-EEGNet. When performing further testing of optimal hyperparameters, we found similar accuracy metrics to CV-EEGNet's paper, finding very similar class imbalances through the Confusion Matrices. However, we wanted to write about the novel part of our project as described above, and hence did not include full metrics for CV-EEGNet model comparisons.

Acknowledgments and Tool Disclosure

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